Recitation 5

Yuchao Chen, Mingtao Huang

{cyc20,huangmt19}@mails.tsinghua.edu.cn

Tsinghua University

March 24/25/26, 2021

Outline

1. **Homework**

2. **Review**

3. **Examples**

4. **Recitation**

Outline

1. **Homework**

2. **Review**

3. **Examples**

4. **Recitation**

1. An internet service provider uses 50 modems to serve the needs of 1000 customers. It is estimated that at a given time, each customer will need a connection with probability 0.01, independent of the other customers.

(a). What is the PMF of the number of modems in use at the given time?

1. An internet service provider uses 50 modems to serve the needs of 1000 customers. It is estimated that at a given time, each customer will need a connection with probability 0.01, independent of the other customers.

(a). What is the PMF of the number of modems in use at the given time?

Note that the probability of $X = 50$ should be dealt with separately. It is the same as the probability that 50 or more out of 1000 customers need a connection:

$$p_X(50) = \sum_{k=50}^{1000} \binom{1000}{k} 0.01^k 0.99^{1000-k}$$

1. An internet service provider uses 50 modems to serve the needs of 1000 customers. It is estimated that at a given time, each customer will need a connection with probability 0.01, independent of the other customers.

(b). Repeat part (a) by approximating the PMF of the number of customers that need a connection with a Poisson PMF.

By approximating the binomial in (a) with a Poisson with parameter $\lambda = 1000 \cdot 0.01 = 10$, we have

$$p_X(k) = e^{-10}\frac{10^k}{k!}, \quad k = 0, 1, \ldots, 49,$$

$$p_X(50) = \sum_{k=50}^{1000} e^{-10}\frac{10^k}{k!}.$$

Or, if you notice that this problem has changed the random variable to the number of "customers", you will get

$$p_X(k) = e^{-10}\frac{10^k}{k!}, \quad k = 0, 1, \ldots, 1000.$$

4. Let X be a RV such that

$$\begin{cases} p_X(1) = p \\ p_X(-1) = 1 - p \end{cases}$$

Find $c \neq 1$, such that $\mathbf{E}[c^X] = 1$.

4. Let X be a RV such that

$$\begin{cases} p_X(1) = p \\ p_X(-1) = 1 - p \end{cases}$$

Find $c \neq 1$, such that $\mathbf{E}[c^X] = 1$.

$$\mathbf{E}[c^X] = \sum_x c^x p_X(x) = cp + c^{-1}(1-p) = 1 \Rightarrow pc^2 - c + 1 - p = 0$$

This can be factorized to $(pc - 1 + p)(c - 1) = 0$.
If you use the formula for quadratic equations, you should further simplify the result:

$$c = \frac{1 \pm \sqrt{1 - 4p(1-p)}}{2p} = \frac{1 \pm \sqrt{(2p-1)^2}}{2p} = \frac{1 \pm (2p-1)}{2p}$$

Outline

1. Homework
2. **Review**
3. Examples
4. Recitation

Review

- Conditional PMF

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}$$

- Conditional expectation

$$\mathbf{E}[X \mid A] = \sum_x x p_{X|A}(x).$$

$$\mathbf{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x|y)$$

$$\mathbf{E}[g(X) \mid Y = y] = \sum_x g(x) p_{X|Y}(x|y)$$

- Total expectation theorem

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X \mid A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X \mid A_n]$$

Review

- Multiple discrete RVs

$$p_{X,Y}(x,y) = \mathbf{P}(X = x, Y = y) = \mathbf{P}(\{X(\omega) = x\} \cap \{Y(\omega) = y\})$$

- Properties of PMF:
  - $\sum_x \sum_y p_{X,Y}(x,y) = 1$
  - $p_X(x) = \sum_y p_{X,Y}(x,y)$
  - $p_{X|Y}(x \mid y) = \mathbf{P}(X = x \mid Y = y) = \dfrac{p_{X,Y}(x,y)}{p_Y(y)}$
  - $\sum_x p_{X|Y}(x \mid y) = 1$
- Expectation:

$$\mathbf{E}[g(X,Y)] = \sum_x \sum_y g(x,y) p_{X,Y}(x,y)$$

$$\mathbf{E}[X + Y + Z] = \mathbf{E}[X] + \mathbf{E}[Y] + \mathbf{E}[Z]$$

Review

- Independence

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \qquad \forall x,y$$

- Conditional indepencence

$$p_{X,Y|A}(x,y) = p_{X|A}(x) \cdot p_{Y|A}(y) \qquad \forall x,y$$

- If X and Y are independent:

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[(h(Y)]$$

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$$

- Memoryless property for Geometric PMF:
  - Given that $X > 1$, the random variable $X-1$ has the same geometric PMF as X.

$$p_{X|X>1}(k) = \frac{\mathbf{P}(\{X > 1\} \cap \{X = k\})}{\mathbf{P}(X > 1)} = \frac{(1-p)^{k-1}p}{1-p}$$
$$= (1-p)^{k-2}p, \qquad k = 2, 3, \ldots$$

$$p_{X-1|X>1}(k) = p_{X|X>1}(k+1)$$
$$= (1-p)^{k-1}p = p_X(k), \qquad k = 1, 2, \ldots$$

  - If the first trial fails, we are back where we started.

Review

- Memoryless property for Geometric PMF:
  - Given that $X > 1$, the conditional expectation of $X-1$ and $(X-1)^2$ are the same as the expectation of X and X^2.

$$\mathbf{E}[X-1|X>1] = \sum_{k=1}^{\infty} k p_{X-1|X>1}(k) = \sum_{k=1}^{\infty} k p_X(k) = \mathbf{E}[X]$$

$$\mathbf{E}[(X-1)^2|X>1] = \sum_{k=1}^{\infty} k^2 p_{X-1|X>1}(k) = \sum_{k=1}^{\infty} k^2 p_X(k) = \mathbf{E}[X^2]$$

  - The conditional expectation of X given $X > 1$ becomes

$$\begin{aligned}
\mathbf{E}[X|X>1] &= 1 + \mathbf{E}[X-1|X>1] = 1 + \mathbf{E}[X] \\
\mathbf{E}[X^2|X>1] &= \mathbf{E}[((X-1)+1)^2|X>1] \\
&= \mathbf{E}[(X-1)^2 + 2(X-1) + 1|X>1] \\
&= \mathbf{E}[(X-1)^2|X>1] + 2\mathbf{E}[X-1|X>1] + 1 \\
&= \mathbf{E}[X^2] + 2\mathbf{E}[X] + 1
\end{aligned}$$

- Memoryless property for Geometric PMF:
  - Mean of Geometric PMF:

$$\begin{aligned}\mathbf{E}[X] &= \mathbf{P}(X=1)\mathbf{E}[X \mid X=1] + \mathbf{P}(X>1)\mathbf{E}[X \mid X>1] \\ &= p + (1-p)(1+\mathbf{E}[X])\end{aligned}$$

$$\mathbf{E}[X] = \frac{1}{p}$$

  - Variance of Geometric PMF:

$$\begin{aligned}\mathbf{E}[X^2] &= \mathbf{P}(X=1)\mathbf{E}[X^2 \mid X=1] + \mathbf{P}(X>1)\mathbf{E}[X^2 \mid X>1] \\ &= p + (1-p)\mathbf{E}[(1+X)^2] = p + (1-p)(1+2\mathbf{E}[X]+\mathbf{E}[X^2])\end{aligned}$$

$$\mathbf{E}[X^2] = \frac{1+2(1-p)\mathbf{E}[X]}{p} = \frac{2-p}{p^2}$$

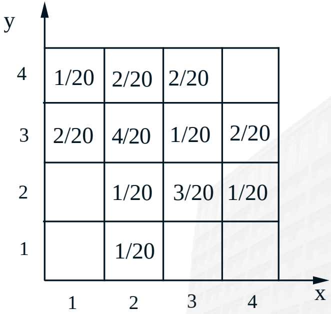
$$\mathrm{var}(X) = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{1-p}{p^2}$$

Outline

1. Homework
2. Review
3. **Examples**
4. Recitation

Marginal and Conditional PMFs

- $p_X(x) = ?$
- $p_Y(y) = ?$
- $p_{X|Y}(x|2) = ?$



Marginal and Conditional PMFs

- $p_X(x) = \sum_y p_{X,Y}(x,y)$
- $p_Y(y) = \sum_x p_{X,Y}(x,y)$
- $p_{X|Y}(x|2) = \dfrac{p_{X,Y}(x,2)}{p_Y(2)}$

y \ x	1	2	3	4
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		

The hat problem

- n people throw their hats in a box and then pick one at random.
- X: number of people who get their own hat
- Find $\mathbf{E}[X]$ and $\mathrm{var}(X)$

The hat problem

- n people throw their hats in a box and then pick one at random.
- X: number of people who get their own hat
- Find $\mathbf{E}[X]$ and $\text{var}(X)$

Solution:

- Define:

$$X_i = \begin{cases} 1, & \text{if } i \text{ selects own hat} \\ 0, & \text{otherwise.} \end{cases}$$

- $X = X_1 + X_2 + \cdots + X_n$
- $\mathbf{E}[X_i] = \dfrac{1}{n}$
- Are the X_i independent? No!
- $\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n] = 1$

The hat problem

- $\mathrm{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \mathbf{E}[X^2] - 1$

$$X^2 = \sum_i X_i^2 + \sum_{i,j:i\neq j} X_i X_j$$

- $\mathbf{E}[X_i^2] = \dfrac{1}{n}$
- $\mathbf{P}(X_1 X_2 = 1) = \mathbf{P}(X_1 = 1) \cdot \mathbf{P}(X_2 = 1 \mid X_1 = 1) = \dfrac{1}{n} \cdot \dfrac{1}{n-1}$
- $\mathbf{E}[X^2] = n \cdot \dfrac{1}{n} + n(n-1) \cdot \dfrac{1}{n(n-1)} = 2$
- $\mathrm{var}(X) = 1$

Outline

1. **Homework**

2. **Review**

3. **Examples**

4. **Recitation**

1. Consider an experiment in which a fair four-sided die (with faces labeled 0, 1, 2, 3) is thrown once to determine how many times a fair coin is to be flipped. In the sample space of this experiment, random variables N and K are defined by

- $N =$ the result of the die roll
- $K =$ the total number of heads resulting from the coin flips

(a) Determine and sketch $p_N(n)$

(b) Determine and tabulate $p_{N,K}(n,k)$

(c) Determine and sketch $p_{K|N}(k \mid 2)$

(d) Determine and sketch $p_{N|K}(n \mid 2)$

(a) Determine and sketch $p_N(n)$

$$p_N(n) = \begin{cases} 1/4, & n = 0, 1, 2, 3 \\ 0, & \text{otherwise.} \end{cases}$$

(b) Determine and tabulate $p_{N,K}(n, k)$

When $N = 0$, the coin is not flipped at all, so $K = 0$. When $N = n$ for $n \in \{1, 2, 3\}$, the coin is flipped n times, resulting in K with a distribution that is conditionally binomial. The binomial probabilities are all multiplied by $1/4$ because $p_N(n) = 1/4$ for $n \in \{0, 1, 2, 3\}$.

(c) Determine and sketch $p_{K|N}(k \mid 2)$

Conditional on $N = 2$, K is a binomial random variable. So we immediately see that

$$p_{K|N}(k|2) = \begin{cases} 1/4, & k = 0, \\ 1/2, & k = 1, \\ 1/4, & k = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(d) Determine and sketch $p_{N|K}(n \mid 2)$

There must have been at least 2 coin tosses, so only $N = 2$ and $N = 3$ have positive conditional probability given $K = 2$.

$$p_{N|K}(2|2) = \frac{\mathbf{P}(\{N = 2\} \cap \{K = 2\})}{\mathbf{P}(\{K = 2\})} = 2/5.$$

Similarly, $p_{N|K}(3|2) = 3/5$.

2. You write a software program over and over, and each time there is probability p that it works correctly, independent of previous attempts. What is the variance of X, the number of tries until the program works correctly?

2. You write a software program over and over, and each time there is probability p that it works correctly, independent of previous attempts. What is the variance of X, the number of tries until the program works correctly?

We recognize X as a geometric random variable with PMF

$$p_X(k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots .$$

The variance of X is given by

$$\text{var}(X) = \sum_{k=1}^{\infty}(k - \mathbf{E}[X])^2(1-p)^{k-1}p.$$

By considering if the first try is successful, we can have

$$\mathbf{E}[X|X>1] = 1 + \mathbf{E}[X].$$

Thus,

$$\mathbf{E}[X] = \mathbf{P}(X = 1)\mathbf{E}[X|X = 1] + \mathbf{P}(X > 1)\mathbf{E}[X|X > 1]$$

from which we obtain $\mathbf{E}[X] = 1/p$. Similarly, we also have

$$\mathbf{E}[X^2|X = 1] = 1, \quad \mathbf{E}[X^2|X > 1] = \mathbf{E}[(1 + X)^2],$$

so that

$$\mathbf{E}[X^2] = p \cdot 1 + (1 - p)(1 + 2\mathbf{E}[X] + \mathbf{E}[X^2]),$$

from which we obtain

$$\mathbf{E}[X^2] = \frac{1 + 2(1 - p)\mathbf{E}[X]}{p} = \frac{2}{p^2} - \frac{1}{p}.$$

We further have $\text{var}(X) = (1 - p)/p^2$.

3. A coin that has probability of heads equal to p is tossed successively and independently until a head comes twice in a row or a tail comes twice in a row. Find the expected value of the number of tosses.

3. A coin that has probability of heads equal to p is tossed successively and independently until a head comes twice in a row or a tail comes twice in a row. Find the expected value of the number of tosses.

Let H_k (or T_k) be the event that a head (or a tail, respectively) comes at the k-th toss, and let p (respectively, q) be the probability of H_k (respectively, T_k). Since H_1 and T_1 form a partition of the sample space, we have

$$\mathbf{E}[X] = p\mathbf{E}[X|H_1] + q\mathbf{E}[X|T_1].$$

Using again the total expectation theorem, we have

$$\mathbf{E}[X|H_1] = p\mathbf{E}[X|H_1 \cap H_2] + q\mathbf{E}[X|H_1 \cap T_2] = 2p + q(1 + \mathbf{E}[X|T_1]).$$

Similarly, we obtain

$$\mathbf{E}[X|T_1] = 2q + p(1 + \mathbf{E}[X|H_1]).$$

Combining the above two relations, collecting terms, and using the fact $p + q = 1$, we obtain after some calculation

$$\mathbf{E}[X|T_1] = \frac{2 + p^2}{1 - pq}, \quad \mathbf{E}[X|H_1] = \frac{2 + q^2}{1 - pq}.$$

Thus,

$$\mathbf{E}[X] = p \cdot \frac{2 + q^2}{1 - pq} + q \cdot \frac{2 + p^2}{1 - pq} = \frac{2 + pq}{1 - pq}.$$